

Certified Perceptual Shielding for Adaptive Bitrate Streaming

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Abstract

Client-side adaptive bitrate (ABR) controllers—whether heuristic, model-based, or learned—share a per-chunk limitation. A request can become unsafe when throughput fluctuates, or wasteful once the rate–quality ladder has saturated in Video Multimethod Assessment Fusion (VMAF). We address both failure modes with a *Certified Perceptual Shield* (CPS). CPS is a model-agnostic runtime projection built from three components. *VMAF-knee bandwidth banking* lowers even feasible proposals to the smallest **rung** whose VMAF lies within a perceptual **budget ε** . The mechanism exploits saturation to bank buffer headroom with negligible quality loss. An online *conformal* lower bound on throughput is calibrated with a finite-sample **$(n+1)$ rank correction**. **Under exchangeability, this bound yields distribution-free coverage $1-\alpha$** . *Certified feasibility* then rejects any download whose time under that bound would exceed the buffer margin. The result is a per-chunk *probabilistic* stall certificate under the conformal assumption. The certificate is a coverage guarantee, not a promise of zero stalls. We evaluate CPS on 204 paired synthetic 5G episodes over twelve reference titles. The titles were chosen for *ladder diversity*, from saturation-rich animation to steep natural motion. Against a conformal safety-only baseline, banking significantly reduces delivered bitrate and rebuffering on every deterministic host we test (greedy through RobustMPC: $-4.5\%/-9.3\%$ to $-3.8\%/-9.2\%$). Mean VMAF differences lie within ± 1.0 (**TOST**). Empirical coverage is 0.919 against **a 0.90 target**. Because effect sizes scale with top-rung saturation, the diversified roster yields conservative but credible savings. A risk-

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aware, predictive variant increases banking ahead of forecast throughput dips (greedy: -14.6% bitrate, -12.6% rebuffering). On source-disjoint real-measured broadband traces (FCC and Norway/Nor-net), banking collapses (-0.9% bitrate). Conformal safety still reduces stalls (-98.7% vs. raw). The payoff is therefore regime-dependent. Optional **PPO** co-training with CPS is reported only as a supporting, single-seed study in the evaluation—not as a headline claim.

Keywords: Adaptive Bitrate Streaming, Runtime Shielding, VMAF, Conformal Prediction, Certified Safety, Policy–Shield Co-design, Video QoE

1. Introduction

Client-side adaptive bitrate (ABR) policies optimize average quality of experience (QoE) through buffer heuristics [1], model-predictive control [2], and deep reinforcement learning (DRL) [3, 4]. Session averages, however, can obscure transient risks associated with individual chunks. A single segment can still stall under volatile sub-6 GHz and 5G/mmWave throughput [5–7], whereas a comfortable buffer invites the opposite inefficiency. Commercial rate ladders often saturate in Video Multimethod Assessment Fusion (VMAF) [8]: further bitrate increases at the top often provide only marginal perceptual gains. Requesting those rungs when the buffer is healthy consumes bandwidth that could cushion the next dip.

A runtime shield sits between the base policy and the player, projecting each proposal before download [9]. The standard repair decrements the representation index until a buffer-feasibility test passes; this strategy is too rigid to adapt effectively. That repair is content-blind: it ignores VMAF saturation, never banks bandwidth proactively, and reduces throughput pessimism to a hand-tuned factor with no coverage guarantee. Deployed ladders carry exploitable structure—rate-quality saturation near the top. That structure supports perceptually negligible bitrate reductions while freeing additional buffer capacity.

We introduce a *Certified Perceptual Shield* (CPS) that targets this structure while stating a formal coverage guarantee:

1. **VMAF-knee bandwidth banking.** Even a feasible action is lowered to the smallest rung j whose VMAF lies within a budget ε of the proposal. The downshift is perceptually bounded, and reclaimed bytes bank as buffer for later chunks.
2. **Conformal throughput lower bound.** Download-time tests use an online split-conformal lower bound on throughput with **target coverage $1-\alpha$** . The

bound is calibrated from residual ratios with a finite-sample $(n+1)$ rank correction [10, 11].

3. **Certified feasibility.** After banking, CPS admits no rung whose download time under the lower bound exceeds the buffer minus a safety margin. This rule yields a per-chunk probabilistic stall certificate with coverage $1-\alpha$ under residual exchangeability, rather than a deterministic no-stall guarantee (~~Proposition 1~~).

Prior VMAF-aware ranking shields do not improve on this collapse on monotone ladders. Maximizing retained VMAF among feasible indices reduces to highest-feasible-index projection (Proposition 2). A perceptual-loss budget therefore cannot constrain choices when VMAF is nondecreasing in the rung index. The wrapper is host-agnostic: the same projection attaches to bitrate-greedy, buffer-based (BBA), and learned PPO controllers alike. To separate the two mechanisms, we run three paired arms on identical episode seeds: RAW (unshielded), SAFETY (conformal feasibility only), and CERTIFIED (feasibility plus banking). Banking then isolates to the CERTIFIED–SAFETY contrast, and safety to SAFETY/CERTIFIED vs. RAW.

We evaluate these claims on 204 paired synthetic 5G episodes over twelve titles selected for ladder diversity, with $\varepsilon=1.0$ and $\alpha=0.10$. Certified banking against safety-only significantly reduces mean bitrate and rebuffering on greedy, BBA, BOLA, and RobustMPC (Table 5). For greedy, the paired differences are $-4.5\%/-9.3\%$ (paired Wilcoxon $p=8.5\times 10^{-4}$). Magnitudes cluster near $\pm 4\%/\pm 9\%$ because steep, non-saturated ladders leave fewer near-lossless downshifts, yet the direction holds everywhere. Mean VMAF differences remain within $\pm\varepsilon$ by TOST equivalence. Empirical conformal coverage is 0.919 against a 0.90 target. A risk-aware, predictive extension widens the perceptual budget when the buffer—or a short look-ahead under a throughput floor—enters a danger zone. On greedy, this variant yields -14.6% bitrate and -12.6% rebuffering. On source-disjoint real-measured broadband traces (FCC and Norway/Nor-net), banking headroom collapses (-0.9% bitrate). Safety still reduces stalls (-98.7% vs. raw greedy). These traces therefore indicate when perceptual banking pays off. Optional PPO co-training with CPS is deferred to a supporting, single-seed study (Section 5.4).

The main contributions are:

- **Certified Perceptual Shield.** A model-agnostic runtime projection that combines VMAF-knee bandwidth banking, a conformal throughput lower bound with finite-sample coverage, and certified buffer feasibility (~~Section 4~~).
- **Isolated banking evaluation.** Paired RAW/SAFETY/CERTIFIED rollouts on

greedy, BBA, BOLA, RobustMPC, and PPO hosts. We report Wilcoxon tests on bitrate and rebuffering, and TOST non-inferiority on VMAF within $\pm\epsilon$ (~~Section 5~~).

- **Risk-aware, predictive banking.** A perceptual budget conditioned on the buffer and a short look-ahead that pre-banks before forecast dips. It improves the trade-off between bandwidth and stalls under the same ϵ ceiling at comfortable buffers (~~Section 4.3~~).
- **Regime characterization.** Banking gains concentrate where top rungs are often feasible (synthetic 5G with a moderate buffer). Real-trace broadband stress is dominated by safety projection (~~Section 5~~).
- **Policy-shield co-design (supporting).** An optional single-seed PPO study with CPS in the training loop; we report the QoE crossover weight w^* but do not treat this arm as a headline claim (~~Section 5.4~~).

The remainder of this paper is organized as follows. Section 2 reviews related work, and Section 3 formalizes the streaming model. Section 4 presents CPS and optional co-training. Section 5 reports the evaluation, Section 6 discusses limitations, and Section 7 concludes.

2. Related work

Three areas bear on client-side representation selection: model-based ABR, learned ABR, and constrained control with runtime shielding. Our focus is the control policy and its runtime action projection; throughput prediction lies outside this scope. Predictors such as CS2P [12] play a complementary role: they feed inputs to model-based or learned controllers [2, 3]. ABR surveys [7, 13, 14] and 5G/mmWave measurements [5, 15] converge on one empirical lesson. Control must remain robust when throughput fluctuates sharply.

2.1. Heuristic and model-based ABR

MPEG Dynamic Adaptive Streaming over HTTP (DASH) separates playout from transport [16]. Clients execute local adaptation policies based on ladder metadata and observed download conditions. Buffer-threshold heuristics such as BBA [1] remain simple and interpretable. BOLA [17], by contrast, derives its decisions from a Lyapunov-based utility formulation. RobustMPC [2] plans ahead. It selects representations by solving a short-horizon model predictive control

(MPC) problem under estimated future throughput. The payoff is interpretable thresholds and horizons. The cost is fragility: performance depends on manual calibration and on prediction accuracy that varies across networks and devices [5, 6]. Moreover, conventional feasibility rules rarely exploit VMAF saturation to bank bandwidth in advance [8]. Adjacent work targets other stages of delivery. SODA [18] pursues smoothness-aware online control, while edge assistance [19] and Oboe-style automatic tuning [20] address separate parts of the system.

2.2. *Deep reinforcement learning for ABR*

Pensieve [3] established the template: a trace-driven DRL formulation for ABR. D-DASH [21] and more recent PPO-based controllers [22] followed this line. Their neural policies combine throughput histories, buffer state, and manifest features [23]. Later systems built on this line. Comyco [24] trains by imitation against a VMAF-based QoE objective and shows that perceptual signals help inside the learned policy. Loki [25] combines learned features with a rule-based model, reducing long-tail failures in production. Fugu [26] deploys learned control in live networks and exposes the distribution shift between simulation and deployment. Taken together, these systems motivate constrained learning and decision-time enforcement. An expected-return objective alone encodes no explicit per-episode stall or smoothness budget [27, 28]. Explicit constraints do not fully close this gap. A controller trained under a CMDP retains guarantees only over the sampled trajectory distribution. An unseen blockage pattern can still induce an unsafe proposal at a single chunk boundary. This gap argues for splitting the stack into a learned performance layer and a runtime enforcement layer. Additional training cannot substitute for runtime validation.

2.3. *Safe/constrained RL, shields, and the content-blindness gap*

CMDPs and Lagrangian primal–dual methods optimize expected return under explicit constraints [29–31]. Safe-RL surveys catalogue related families: shielding, Lyapunov, and risk-sensitive schemes [28, 32]. COREL [27] is closest to our training formulation. It casts ABR over commercial 5G/mmWave traces as a CMDP with a stall-time constraint. The CMDP is solved by a primal–dual actor–critic. That constraint, however, operates within training; it is not a standalone runtime projection with intervention telemetry. Runtime shielding takes the opposite approach: it repairs a proposed action before execution and can log each intervention [9]. Surveys split the family into *pre-shielding*, which prunes the action set, and *post-shielding*, which overwrites an unsafe proposal [33]. By that

taxonomy, CPS is a post-shield. Its safety set is drawn by a conformal feasibility test rather than a temporal-logic model.

This distinction shapes our design: we separate an optional learned performance layer from a deterministic runtime CPS. Learned policies such as Lagrangian PPO can pursue expected stall and smoothness budgets across training rollouts [27, 30]. They cannot certify per-chunk feasibility under conformal throughput bounds. CPS targets exactly this gap. Before every download it (i) banks bandwidth on VMAF-saturated rungs within a perceptual budget ε , and (ii) projects onto actions feasible under a distribution-free throughput lower bound with coverage $1-\alpha$ [9–11]. Its use of perceptual data also differs from Comyco, where VMAF enters only the learned objective [24]. CPS reads the ladder at decision time for proactive banking, rather than only emergency index repair. Table 1 summarizes this positioning.

Table 1: Positioning of client-side ABR methods by training constraints, deterministic runtime projection, conformal throughput bounds, and VMAF-knee banking.

Work	Year	Paradigm	Constr.	Guard	Conf.	Bank
BBA [1]	2014	Heuristic	N	N	N	N
BOLA [17]	2018	Lyapunov util.	Imp.	N	N	N
RobustMPC [2]	2015	MPC/lookahead	N	N	N	N
Pensieve [3]	2017	DRL	N	N	N	N
Comyco [24]	2019	Quality-aware IL	N	N	N	N
Loki [25]	2021	Rule-RL fusion	N	$\approx$	N	N
COREL [27]	2023	Constr. DRL	Y	N	N	N
SODA [18]	2024	Online control	N	N	N	N
Ours (CPS)	—	Shield ($\pm$ RL)	$\approx$	Y	Y	Y

Constr.: explicit stall budget training (Y), implicit (Imp.), optional/partial ($\approx$), or no ($\emptyset$). **Guard:** auditable action projection before download [9].

Conf.: conformal throughput lower bound with stated coverage [10]. **Bank:** VMAF-knee bandwidth banking within a perceptual budget ε .

3. System model and problem formulation

This section formalizes client-side ABR at the chunk timescale. A base policy proposes a ladder index; CPS applies a deterministic projection before download; the player then updates the buffer under realized throughput. The subsections below define the architecture, the state–action–reward tuple, the ladder saturation that banking exploits, and the download and buffer dynamics. CPS internals—the conformal bound, safety set, and knee rule—are deferred to Section 4 ~~so C_k is defined before use.~~ Table 2 collects the symbols used throughout.

Table 2: Summary of notation.

Symbol	Meaning
k, T_c, K_e	chunk index, chunk duration, chunks per episode
$L, \mathcal{B}=\{0, \dots, L-1\}$	number of ladder rungs, rung index set
$a_k^{\text{raw}}, a_k^{\text{exec}}, \iota_k$	proposed rung, executed rung, intervention bit
B_k, m, B_{crit}	buffer level, safety margin, critical buffer
$V_k(j), \sigma_k(j)$	VMAF and size (bits) of rung j at chunk k
$\varepsilon, \varepsilon_{\text{eff}}, \varepsilon_{\text{risk}}$	perceptual budget, effective/risk-mode budget
$\hat{C}_k, C_k^{\text{act}}, \underline{C}_k$	throughput predictor, realized, conformal lower bound
$\alpha, 1-\alpha$	conformal miscoverage, target coverage
$r=C^{\text{act}}/\hat{C}, W, \kappa$	residual ratio, residual window, predictor horizon
$d_k(j), H, \varepsilon_0$	download time of rung j , look-ahead, numerical guard
μ, β	logged-QoE smoothness and stall weights ((2))
w, w^*	QoE stall weight (sweep), mean-QoE crossover weight

3.1. Streaming architecture and control loop

We work in the standard Hypertext Transfer Protocol (HTTP) adaptive-streaming abstraction [2, 3, 16]. A video origin or content delivery network (CDN) publishes L encoded representations. We index them by $\mathcal{B} = \{0, \dots, L-1\}$; the DASH/HTTP Live Streaming (HLS) manifest carries the matching per-chunk sizes and perceptual metadata. At decision step k , the client fetches chunk k while holding a playout buffer B_k . The effective trace throughput $\hat{C}_k$ is exogenous to the controller. This assumption reflects deployment conditions rather than merely simplifying the model. On sub-6 GHz and 5G/mmWave access, throughput can change abruptly because of blockage, **LOS/NLOS** transitions, scheduling, and mobility [5, 6].

A base policy π (heuristic or learned) maps an augmented state s_k to a proposed action $a_k^{\text{raw}} \in \mathcal{B}$. Before chunk k is downloaded, the Certified Perceptual Shield $\mathcal{S}$ applies the deterministic projection

$$(a_k^{\text{exec}}, \iota_k) = \mathcal{S}(s_k, a_k^{\text{raw}}), \quad a_k^{\text{exec}} \in \mathcal{B}, \quad \iota_k \in \{0, 1\}, \quad (1)$$

The projection returns the executed representation index a_k^{exec} and an audit bit ι_k , the latter indicating whether the projection altered the proposal. Only a_k^{exec} feeds the chunk download and buffer dynamics (4); Fig. 1 traces the $a_k^{\text{raw}} \rightarrow \mathcal{S} \rightarrow a_k^{\text{exec}}$ path.

Decisions advance on the chunk clock, one step per playback duration T_c (4 s in our harness) [2, 3].

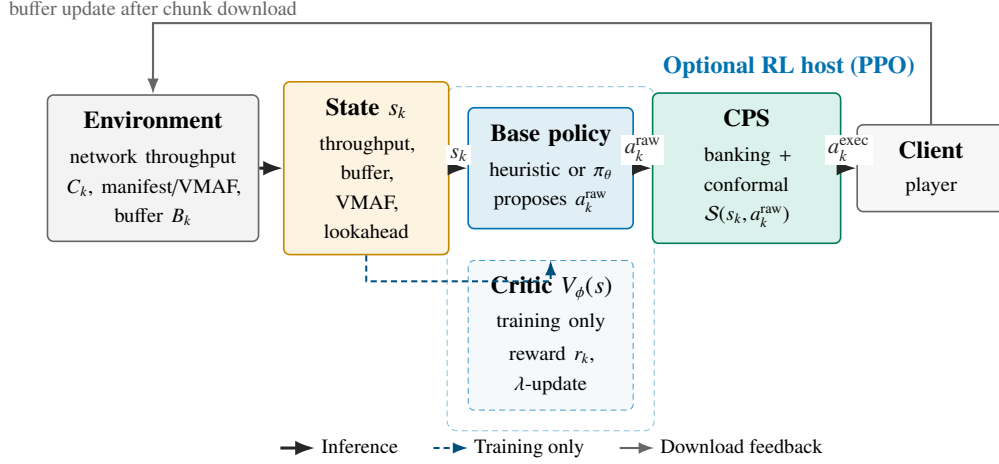


Figure 1: Control loop. The base policy proposes a_k^{raw} ; CPS banks on the VMAF knee and projects under a conformal throughput lower bound to produce a_k^{exec} . Training-only dashed paths (critic, dual variables) are optional for RL hosts.

The two layers serve distinct roles. Optional learned-host training (Section 3.5) handles stall and smoothness budgets in expectation over rollouts [30, 31]. However, it does not guarantee per-step feasibility under conformal throughput bounds. CPS provides this guarantee through decision-time banking and certified repair [9, 10].

3.2. State, action, reward, and nonlinear perceptual mapping

State s_k . The observation holds buffer occupancy, past throughput samples, and the previous decision. It also includes manifest lookahead: upcoming chunk sizes across the representation ladder, plus spatial/temporal complexity tags (SI/TI) [2, 3, 16]. The VMAF ladder $\{V_k(j)\}_{j \in \mathcal{B}}$ completes the observation, where $V_k(j)$ is the perceptual score for chunk k at representation j [8]. This last component is essential: without $\{V_k(j)\}$, neither a learned host nor CPS can detect the content-dependent ladder saturation that bandwidth banking exploits.

Nonlinear, content-dependent fidelity. Saturation arises because quality does not scale linearly with rate: for each chunk k , the map $j \mapsto V_k(j)$ is not affine in the discrete index j . This map also varies across chunks. Different group-of-pictures (GOP) and texture structures yield different marginal VMAF gains between adjacent rungs. Therefore, the bitrate-to-VMAF relationship is both nonlinear and content-dependent. Consequently, adjacent rungs may differ by a few VMAF points or by more than ten. Quality often saturates at the upper rungs of the ladder—precisely the structure that CPS banking exploits [8]. Perceptually

aware, JND-based per-title encoding aims to remove this saturation at authoring time [34]. In contrast, CPS exploits any saturation that remains in a deployed ladder at decision time.

Action. $a_k^{\text{raw}} \in \mathcal{B}$ is the policy’s proposed representation index for chunk k ; the environment executes only a_k^{exec} .

QoE surrogate. Keeping to RobustMPC/Pensieve reporting conventions [2, 3], we log a scalar per-chunk surrogate with fidelity $q_k \equiv V_k(a_k^{\text{exec}})$:

$$\text{QoE}_k^{(\text{rep})} = q_k - \mu \text{smooth}_k - \beta \tau_k^{\text{stall}}, \quad (2)$$

where $\text{smooth}_k = |q_k - q_{k-1}|$ is the absolute VMAF change between adjacent chunks and τ_k^{stall} is the stall duration at step k . The first chunk takes a zero smoothness term. Our shared harness fixes $\beta=4.3$ per stall second and $\mu=1.0$ per absolute VMAF-point change. The symbol α remains reserved for the conformal miscoverage throughout. The rebuffering weight is a Pensieve-style penalty, rescaled onto the VMAF fidelity axis. Session QoE sums over chunks: $\sum_{k \in \mathcal{K}_e} \text{QoE}_k^{(\text{rep})}$ (Section 5). For a learned host, optional RL training substitutes a Lagrangian CMDP objective over rollouts (Appendix A) [29, 30].

3.3. Ladder saturation

The evaluated system reports one VMAF value per representation and video (pooled per-frame scores), so shield decisions operate on a session-mean ladder that does not vary with chunk index k . Table 3 summarizes session-mean ladders for 12 publicly available reference titles (six Blender animations, six Xiph DERF clips); Fig. 2 illustrates per-chunk spread on a legacy four-title subset where frame-level VMAF logs were retained. Across the twelve-title suite, 67% exhibit a non-monotone session-mean ladder. The mean top-rung gain (6000 vs. 2850 kbps) is only 5.61 VMAF points. This pattern reflects strong saturation on animation titles (e.g. Big Buck Bunny at +0.03) but steeper climbs on natural content (e.g. Park Joy at +14.3). Two properties make banking worthwhile. First, many adjacent pairs differ by at most one VMAF point on saturated titles, so a single-step downgrade costs almost nothing perceptually. Second, the top rungs saturate frequently, allowing the knee rule (7) to bank bytes with no material quality loss. For the monotone per-video ladder evaluated here, ranking feasible indices by VMAF produces the same choice as projection to the highest feasible index. CPS instead selects the smallest within-budget rung. This choice takes effect only when saturation renders downshifts near-lossless. Chunk-level inversions appear in Fig. 2 for the logged subset. However, banking does not depend on them. Section 5.5 shows that the

knee rule remains effective. In fact, it banks more when the shield observes the raw per-chunk ladder.

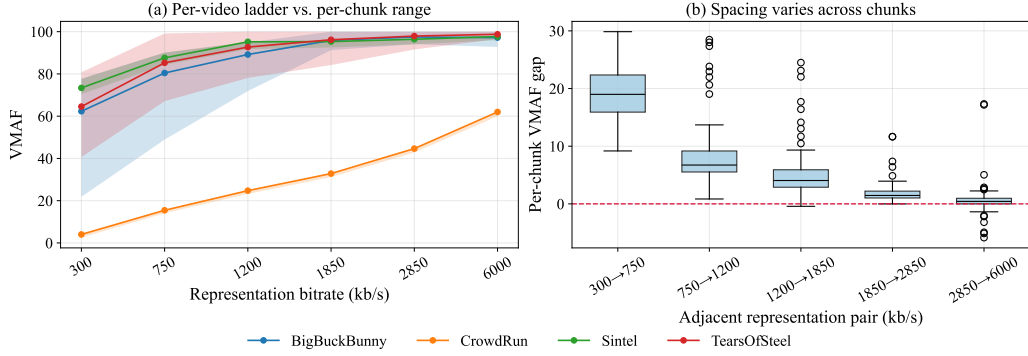


Figure 2: Representation ladder measured from the per-frame VMAF logs. (a) Per-video pooled ladder (markers) against the range spanned by individual chunks (shaded). (b) Distribution of the VMAF gap between adjacent representations across all 173 chunks; the dashed line marks a zero gap, below which the higher representation scores worse.

Table 3: Session-mean representation ladder across twelve evaluation titles.

Video	Held-out	SI	TI	Top gain	Span	Mono.
Big Buck Bunny	—	60.6	47.1	+0.03	30.6	Yes
Tears of Steel	—	41.0	31.8	+4.09	26.1	No
Sintel	✓	50.4	14.7	+1.30	22.9	Yes
Elephants Dream	—	51.8	27.0	+7.87	31.2	No
Ducks Take Off	—	94.9	16.6	+11.59	36.6	No
Cosmos Laundromat	✓	42.6	38.3	+5.23	34.6	No
Park Joy	—	119.7	34.8	+14.31	46.5	No
Old Town Cross	—	61.2	12.6	+6.85	49.5	Yes
Rush Hour	—	27.7	11.7	+4.00	29.8	No
Into Tree	—	47.4	14.4	+9.12	36.1	No
Sunflower	—	38.0	16.5	+1.42	24.9	No
Kristen & Sara	✓	85.8	7.0	+1.53	37.6	Yes

Top gain is $\text{VMAF}(6000) - \text{VMAF}(2850)$ in kbps; Span is $\text{VMAF}(6000) - \text{VMAF}(300)$. Mono. marks a non-decreasing session-mean ladder. Held-out titles (✓) are excluded from PPO training. SI/TI are spatial/temporal complexity indices.

3.4. Chunk-timescale dynamics

3.4.1. Throughput and latency

Let $\hat{C}_k > 0$ be the trace-derived effective throughput, in kb/s, for chunk k . Every method sees the identical throughput trace in paired evaluations. The choice of

a_k^{exec} sets the manifest payload size $\sigma_k(a_k^{\text{exec}})$ in bits. Write $\tilde{C}_k = \max\{\hat{C}_k, \varepsilon_0\}$ for a small constant $\varepsilon_0 > 0$ (the same numerical guard as in (5)). The nominal download duration is $d_k = \sigma_k(a_k^{\text{exec}})/(10^3 \tilde{C}_k)$ seconds [2].

3.4.2. Buffer and stall

Let $B_k \geq 0$ be the buffered playback time when the download of chunk k begins [2, 3]. Rebuffering (stall) and the buffer update follow

$$\tau_k^{\text{stall}} = \max\{0, d_k - B_k\}, \quad (3)$$

$$B_{k+1} = \max\{0, B_k - d_k\} + T_c. \quad (4)$$

Randomness across episodes has a single source: trace and initial-condition sampling [13, 35, 36]. Given $(s_k, a_k^{\text{exec}}, \hat{C}_k)$, each transition is deterministic.

3.5. Optional Lagrangian training for RL hosts

When the base policy is learned, we may train PPO with a Lagrangian-relaxed constrained MDP (CMDP) objective that penalizes rebuffering and smoothness alongside fidelity [29–31]. The dual variables adapt as training proceeds; Appendix A details the optional buffer-deviation and Lyapunov stabilizers [37]. None of this certifies per-chunk feasibility, which finite-sample PPO with clipped objectives cannot provide [4]; CPS supplies this layer at deployment time. Co-design wraps the training environment in CPS, so the actor–critic observes the projected action distribution, whereas shield-at-eval-only policies train without it.

4. Method: Certified Perceptual Shield

CPS is a deterministic, model-agnostic wrapper around any base policy that proposes $a_k^{\text{raw}} \in \mathcal{B}$. The rest of this section covers the conformal estimator, the banking and feasibility projection (~~Algorithm 1~~), a predictive extension, optional PPO co-training, and the evaluation arms.

4.1. Conformal throughput lower bound

The point predictor is the harmonic mean of the last κ observed throughputs, RobustMPC-style [2]. After each step, the multiplicative residual $r = C^{\text{act}}/\hat{C}$ is appended to a rolling window of size W . Given n residuals, the split-conformal α -quantile uses the finite-sample rank $\lfloor \alpha(n+1) \rfloor$ (1-based). A linearly interpolated empirical quantile is excluded: it under-covers on short episodes [10, 11]. If that rank is undefined— n too small—we fall back to a fixed scale $\rho_{\text{fb}}=0.80$. The

lower bound reads $\underline{C}_k = \hat{C}_k \cdot q_\alpha$, clipped to stay at or below $\hat{C}_k$. Under residual exchangeability, the construction guarantees $\mathbb{P}(C^{\text{act}} \geq \underline{C}_k) \geq 1 - \alpha$. Given $\underline{C}_k$, the certified download duration of rung j is

$$d_k(j) = \min \left\{ \frac{\sigma_k(j)}{\underline{C}_k \cdot 10^3 + \varepsilon_0}, 60 \right\}. \quad (5)$$

Proposition 1 (Per-chunk stall certificate). *Fix chunk k , executed index j , buffer B_k , and conformal lower bound $\underline{C}_k$ computed with miscoverage α . If $d_k(j) \leq \max\{B_k - m, 0.1\}$ via (5) and the realized throughput satisfies $C_k^{\text{act}} \geq \underline{C}_k$, then $\tau_k^{\text{stall}} = 0$. Under exchangeability of conformal residuals, $\mathbb{P}(C_k^{\text{act}} \geq \underline{C}_k) \geq 1 - \alpha$ at each step where the bound is defined.*

Proof sketch. When $C_k^{\text{act}} \geq \underline{C}_k$, the realized download time is at most $d_k(j)$, which by assumption does not exceed B_k ; hence (4) gives $\tau_k^{\text{stall}} = 0$. Coverage follows from standard split-conformal prediction with the $(n+1)$ rank correction [10, 11]. $\square$

Scope of the guarantee. The certificate provides marginal coverage over the exchangeable residual window: $\mathbb{P}(C_k^{\text{act}} \geq \underline{C}_k) \geq 1 - \alpha$, averaged over draws. It does not provide coverage conditional on the current buffer or throughput regime. Distribution-free split conformal cannot do so without stronger assumptions [11].

Abrupt 5G/mmWave LOS/NLOS and scheduling shifts can break residual exchangeability [5]. The online defense is a sliding residual window of length W : recent ratios both calibrate q_α and immediately precede the predicted chunk. The bound therefore tracks local nonstationarity, even though finite-sample exactness holds only on an exchangeable window. If drift is large, coverage decays. The decay is bounded by a total-variation term between calibration and test residuals [38]. A weighted-conformal variant that down-weights stale residuals is a natural extension, which we leave to future work. Empirically, we treat reported coverage as the operational check of Proposition 1. ~~On the primary synthetic 5G pool, the certified arm meets $1 - \alpha$ (0.919 vs. 0.90).~~ Real broadband traces and tighter α can sit marginally below the target (Section 5.3). This pattern matches the finite-sample and nonstationarity stress that the theory predicts.

4.2. Banking and certified feasibility

Safety set. Indices that never exceed the proposal and fit the buffer margin m form

$$\mathcal{A}_{\text{safe},k}(a) = \{j \leq a : d_k(j) \leq \max\{B_k - m, 0.1\}\}. \quad (6)$$

VMAF-knee banking. Even when a is feasible, CPS selects the smallest rung within a perceptual budget ε of the proposal,

$$j_k^{\text{knee}}(a) = \min\{j \leq a : V_k(a) - V_k(j) \leq \varepsilon\}, \quad (7)$$

then projects $j_k^{\text{knee}}(a)$ onto $\mathcal{A}_{\text{safe},k}(a)$; safety always takes precedence over ε . **Remark.** Banking may run a lower rung than the proposal, and a lower rung can only shorten $d_k(\cdot)$. Once (6) holds, safety projection therefore keeps the certificate intact.

Banking distinguishes CPS from prior VMAF-ranking shields, and the distinction is substantive. On the monotone ladders seen in practice, a ranking shield provably banks nothing.

Proposition 2 (Ranking shields collapse on monotone ladders). *Fix chunk k and suppose the VMAF ladder is nondecreasing in the rung index, $V_k(0) \leq \dots \leq V_k(L-1)$, and that download time $d_k(\cdot)$ is nondecreasing in the rung index. Let $\mathcal{F} = \{j : d_k(j) \leq B_k - m\}$ be the feasible set. Then any shield that returns $\arg \max_{j \in \mathcal{F}} V_k(j)$ returns the highest feasible index $\max \mathcal{F}$, independently of any perceptual-loss budget ε .*

Proof. Because d_k is nondecreasing, $\mathcal{F}$ is a prefix $\{0, \dots, j_{\max}\}$ with $j_{\max} = \max \mathcal{F}$. Since V_k is nondecreasing, $\arg \max_{j \in \mathcal{F}} V_k(j) = j_{\max}$; the budget ε never enters the maximization. $\square$

Ranking by retained VMAF is therefore equivalent to projection onto the highest feasible index, and it frees no bandwidth. CPS instead uses the proposal's quality and steps down to the smallest rung within ε of it (7). This mechanism lets CPS exploit rate–quality saturation: when top rungs carry near-identical VMAF, the knee lies well below $j_{\max}$, and the freed bytes bank as buffer.

Algorithm 1 realizes (7)–(6). If $B_k \leq B_{\text{crit}}$, CPS forces the lowest rung. Otherwise it computes the knee under the effective budget ε_{eff} (equal to ε unless predictive mode is on; Section 4.3), then decrements the index until (5) respects the margin m . The shield never upgrades a proposal. Diagnostics log banked bits, VMAF surrendered, and whether realized throughput met $\underline{C}_k$ (empirical coverage).

Evaluation arms. RAW: run a^{raw} unshielded. SAFETY: Algorithm 1 with banking switched off ($j^{\text{knee}}=a$), leaving conformal feasibility alone. CERTIFIED: the full CPS, banking plus feasibility. Paired seeds keep the Wilcoxon and TOST tests valid across arms.

Algorithm 1 Certified Perceptual Shield (compact)

1: **Input:** $a^{\text{raw}}, B_k, \{\sigma_k(j), V_k(j)\}$, conformal estimator; parameters $(\varepsilon, m, B_{\text{crit}})$
2: **Output:** (a^{exec}, ι) with $\iota = \mathbf{1}[a^{\text{exec}} \neq a^{\text{raw}}]$
3: $a \leftarrow \text{clip}(a^{\text{raw}}); \underline{C}_k \leftarrow$ conformal lower bound; define $d_k(\cdot)$ via (5)
4: **if** $B_k \leq B_{\text{crit}}$ **then**
5: **return** $(0, \mathbf{1}[a \neq 0])$
6: **end if**
7: compute ε_{eff} (Section 4.3); $j \leftarrow j_k^{\text{knee}}(a)$ via (7) with ε_{eff}
8: **while** $j > 0$ and $d_k(j) > \max\{B_k - m, 0.1\}$ **do**
9: $j \leftarrow j - 1$
10: **end while**
11: $a^{\text{exec}} \leftarrow \min\{a, j\}$; **return** $(a^{\text{exec}}, \mathbf{1}[a^{\text{exec}} \neq a])$

4.3. Predictive and risk-aware perceptual budget

The predictive budget widens when slack is scarce: it banks lightly when buffer headroom is ample and more aggressively when risk arises. While the buffer stays comfortable, $\varepsilon_{\text{eff}} = \varepsilon$ holds quality steady. Two triggers widen the budget. The first is a drop of B_k below a risk threshold B_{risk} . The second is a short H -step look-ahead under a planning throughput floor (a low quantile of recent observations) that forecasts a dip below B_{risk} ,

$$\varepsilon_{\text{eff}} = \varepsilon + (\varepsilon_{\text{risk}} - \varepsilon) \cdot f(B_k, \text{lookahead}), \quad (8)$$

with $f \in [0, 1]$. This lets the shield pre-bank more deeply before stalls force emergency cuts. The “improved” experiments run with $\varepsilon=1.0$, $\varepsilon_{\text{risk}}=4$, $B_{\text{risk}}=8$ s, and $H \geq 6$ at forecast quantile 0.2.

4.4. Optional learned hosts and co-design

CPS does not require a learned policy. When the host is an RL policy, we train PPO (optionally Lagrangian-shaped) with Stable-Baselines 3 [4, 39] on content-aware observations: throughput history, buffer, SI/TI, VMAF ladder. Co-design: wrapping the training environment in CPS changes the action distribution observed by the actor–critic. The policy can then learn to propose bolder rungs that banking will trim for safety. To gauge the effect, we compare two policies against each other under identical 5G train/test provenance: one trained without the shield (shield-at-eval-only), one co-trained with CPS. Both face the improved shield at test time.

4.5. Implementation notes

At the primary 5G operating point the defaults are $\alpha=0.10$ (coverage target 0.90), window $W=200$, $\kappa=5$, margin $m=0.5$ s, $B_{\text{crit}}=0.3$ s, and buffer cap 12 s.

Table 4 collects the shield and predictive-banking defaults from the primary experiments. Empirical coverage is the fraction of steps with $C^{\text{act}} \geq \underline{C}_k$. Per decision, the projection runs in $O(L+H)$ over the L ladder rungs and an H -step look-ahead. Conformal-quantile upkeep on the length- W residual window is $O(1)$ amortized. Across 20,000 rollout decisions on a commodity CPU ($L=6$, full predictive shield), per-decision wall time is $131\ \mu\text{s}$ at the median and $310\ \mu\text{s}$ at the 99th percentile. This is roughly 10^{-5} of the 4 s chunk budget. CPS therefore runs inline, and the control loop is unaffected.

Table 4: CPS default hyperparameters (primary 5G suite).

Parameter	Value
Perceptual budget ε	1.0 VMAF point
Conformal miscoverage α	0.10 (target coverage 0.90)
Residual window W / predictor κ	200 / 5
Buffer margin m / critical B_{crit}	0.5 s / 0.3 s
Risk budget $\varepsilon_{\text{risk}}$ / B_{risk}	4.0 / 8 s
Look-ahead horizon H / forecast quantile	≥ 6 / 0.2

5. Experimental evaluation

We organize the evaluation around six research questions:

RQ1 *Banking vs. safety.* Does VMAF-knee banking reduce bitrate and rebuffering relative to conformal safety-only projection, with VMAF inside $\pm\varepsilon$?

RQ2 *Coverage.* Does empirical conformal coverage meet the finite-sample target $1-\alpha$?

RQ3 *Predictive banking.* Does a risk-aware, look-ahead budget deepen savings without leaving the perceptual envelope?

RQ4 *Co-design.* How does training with CPS in the loop change the certified operating point relative to shield-at-eval-only operation?

RQ5 *Per-chunk robustness.* Does banking remain sound and near-lossless when the shield operates on the true (non-monotone) per-chunk ladder rather than the pooled session-mean ladder?

RQ6 *Hyperparameter sensitivity.* Are the results stable across the perceptual budget ε and miscoverage α , and does coverage track the target $1-\alpha$?

5.1. Experimental setup and datasets

The testbed is a multi-video, chunk-timescale emulator ($K_c=48$ chunks/episode, $T_c=4$ s, buffer cap 12 s unless noted). It drives twelve publicly available reference titles with session-mean VMAF ladders [8]: six Blender open animations (Big Buck Bunny, Tears of Steel, Sintel, Elephants Dream, Ducks Take Off, Cosmos Laundromat) and six Xiph DERF clips (Park Joy, Old Town Cross, Rush Hour, Into Tree, Sunflower, Kristen & Sara). We selected them for ladder structure, not catalogue breadth: saturation-rich animation (Big Buck Bunny, Tears of Steel), steep natural motion (Park Joy, Ducks Take Off), wide-span DERF clips (Into Tree, Old Town Cross), and mixed held-out titles (Sintel, Cosmos Laundromat, Kristen & Sara) for generalization checks (Table 3). Nine titles appear in PPO co-design training; three are held out entirely (Sintel, Cosmos Laundromat, Kristen & Sara). **The negative broadband result (Section 5.3) shows that we do not select regimes that favor banking.** CPS is a deterministic per-chunk projection fixed by the observed VMAF ladder and buffer state, so ladder shape—not semantic content—drives its effect. The twelve-title suite therefore spans both saturated and steep/inverted session-mean ladders (67% non-monotone).

Primary 5G suite. The primary pool comprises synthetic 5G traces with intermittent dips and outages, held out from the co-design training seeds (204 paired episodes). We generate them synthetically because public per-chunk 5G/mmWave throughput logs suitable for chunk-level ABR replay remain scarce. For realism, the generator is tuned to reported 5G/mmWave dip-and-outage dynamics [5, 15], and the broadband suite below adds a real-network counterpart. Train and test trace pools stay source-disjoint. Co-design policies train on the training pool alone and face held-out test seeds.

Broadband suite. The broadband suite uses real-measured traces: the FCC Measuring-Broadband-America fixed-line logs [35] and the Norway/Nor-net mobile-broadband mobility logs [36] (27 held-out test traces, source-disjoint from training). Both are widely used public datasets in ABR research. They provide a real-network counterpart to the synthetic 5G suite and show where banking offers limited headroom.

Hosts. Five host families cover the ABR design space: bitrate-greedy (always at the top rung), buffer-based ABR (BBA-like), BOLA (the Lyapunov buffer-based rule of the dash.js reference player [17, 40]), robust model-predictive control (RobustMPC [2]), and PPO policies (content-aware “proposed” and content-blind Pensieve-style). RobustMPC is a five-segment planner over a throughput prediction discounted by recent worst-case error. CPS carries no host bias: one wrapper, applied at inference across all five families, tuned to none of them.

Arms and pairing. Every host rolls out RAW, SAFETY, and CERTIFIED on identical Python seeds—the same video–trace pairs. We report mean delivered bitrate, total rebuffering, stall-free rate, mean VMAF, intervention rate, and empirical conformal coverage. Significance rests on paired Wilcoxon signed-rank tests (zero-method `zsplit`) and TOST equivalence of VMAF within $\pm\epsilon=1.0$ [41]. Defaults are $\alpha=0.10$ and coverage target 0.90. We set $\epsilon=1.0$ as both the knee budget and the TOST margin. One VMAF point is a small fraction of typical adjacent-rung and top-rung gaps on our ladders (Table 3) and is conservative relative to perceptually aware per-title encoding practice [34]. It is an engineering design choice aligned with (7), not a preregistered clinical equivalence margin [41]. **The p -values we quote are raw and two-sided.** Since several paired contrasts run at once (bitrate and rebuffering across the host families and two shield settings), we checked that every reported bitrate and rebuffering effect survives Holm–Bonferroni correction at 0.05 over the reported family. The paired-episode bootstrap additionally supports the co-design VMAF and rebuffering contrasts. A caveat on diversity follows. Each episode is one (video, trace) pair—video drawn uniformly from the twelve titles, trace from the held-out pool. Observations therefore cluster by title, and effective independent-content diversity is twelve, not 204.

We treat the paired tests as evidence of within-pair shield effects—the shielded arm versus the unshielded arm on the same video–trace pair—and we report per-title ladder structure (~~Table 3~~) with held-out breakdown where applicable. Vanishingly small p -values (say $< 10^{-4}$) track sign-consistency of a paired difference over 204 pairs. They constitute strong evidence for a within-pair effect, not precise magnitudes.

5.2. Model-agnostic banking on 5G (RQ1–RQ2)

Table 5 lays out absolute paired-arm metrics on synthetic 5G test traces for five host families: greedy, BBA, BOLA, RobustMPC, and a Pensieve-style negative control. Figure 3 then distills the CERTIFIED vs. SAFETY contrast across hosts and regimes. Banking behaves consistently across all four deterministic hosts on the twelve-title suite. **Certified–safety** bitrate reductions are -4.5% , -4.1% , -4.1% , and -3.8% , with rebuffering cuts of -9.3% , -8.9% , -8.9% , and -9.2% (Table 5; greedy $p=8.5 \times 10^{-4}$). Steep and non-saturated ladders (Table 3) keep absolute savings modest—the saturation-dependent regime the mechanism predicts—yet every host stays perceptually near-lossless. Partitioning the greedy certified–safety contrast by session-mean top-rung VMAF gain (Table 3) makes that dependence explicit. On titles with top gain at most 2 VMAF points (Big Buck Bunny, Sintel, Sunflower, Kristen & Sara; $n=60$ episodes) banking saves 4.6% bitrate and 9.9%

rebuffering. On steep titles with top gain at least 9 (Park Joy, Ducks Take Off, Into Tree; $n=51$) the same contrast shrinks to 2.1%/3.7%. The pooled headline (-4.5% / -9.3%) therefore mixes both regimes rather than reflecting a saturation-only subset. Mean VMAF differences are perceptually negligible, at -0.01 , 0.17 , 0.18 , and 0.02 points across greedy, BBA, BOLA, and RobustMPC, with every value inside $\pm\epsilon$ by TOST. The two shield layers split the work. Against RAW, certified greedy cuts rebuffering by -55.5% (BBA: -51.0%). Conformal safety eliminates the catastrophic stalls; banking adds the rest of the headroom. Figures 5 and 6 show per-episode operating points and rebuffering CDFs for greedy and BBA. Coverage follows a distinct pattern on the synthetic 5G pool. Empirical coverage there is 0.919 (greedy) and 0.919 (BBA) against a 0.90 target, consistent with the $(n+1)$ conformal calibration at these episode lengths (Section 4.1). Because coverage depends on the throughput trace and the conformal calibrator rather than the base policy, it is identical (≈ 0.919) across every host and shield variant on the 5G test set (Fig. 4). Proposition 1 is a per-chunk probabilistic certificate, not a per-episode stall-free one. At coverage 0.919 , about 8% of chunks can have realized throughput fall below $\underline{C}_k$. The forced downshift at B_{crit} and the first-chunk cold start explain the residual rebuffering in the certified arm. Pensieve, the negative control, degrades to near-floor bitrates with near-zero stalls, so CPS rarely intervenes. Banking cannot rescue a policy that never requests saturated rungs.

That high intervention rate under greedy (Table 5) is fixed by construction; it does not imply that the shield dominates an informed base policy. Greedy proposes the top rung every time, so nearly every banking downshift or feasibility repair appears as an intervention. The rate gauges how far a host sits from the certified operating point, not how much of the outcome to attribute to the shield. This is why we run the model-agnostic contrast across hosts spanning the spectrum.

Certified intervention decreases across all hosts: 70.4% for always-top greedy, 61.1% for BOLA, 31.1% for RobustMPC—which already plans conservatively against a discounted throughput estimate—and near zero for the conservative Pensieve-style control. The sharper the base policy, the lighter the shield’s load. Coverage nevertheless holds at 0.919 throughout. The certified metrics therefore depend increasingly on the base policy’s own decisions, with intervention only where banking headroom remains.

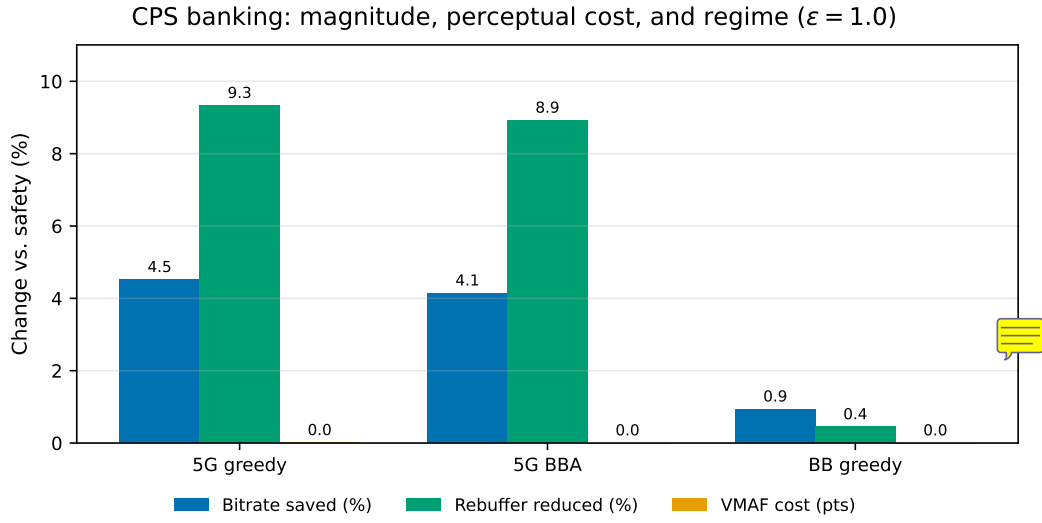


Figure 3: CPS banking overview (certified vs. safety, 204 paired episodes, $\varepsilon=1.0$). For two synthetic 5G hosts and a broadband host, bars show percent bitrate saved, percent rebuffering reduced, and mean VMAF cost (points). Banking is significant and perceptually near-lossless on 5G (magnitude scales with ladder saturation) but collapses on broadband, where top rungs are rarely jointly feasible and saturated.

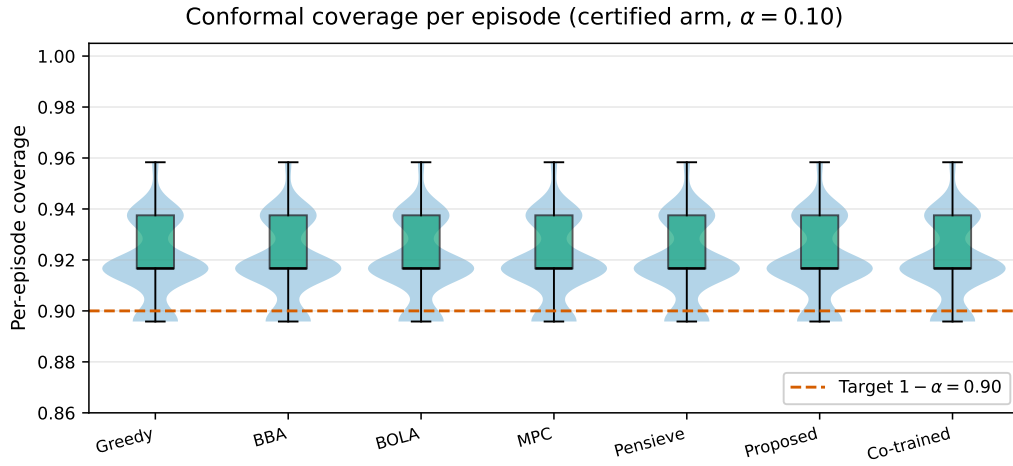


Figure 4: Per-episode conformal coverage of the certified arm on the synthetic 5G test pool ($\alpha=0.10$, target 0.90, dashed). Coverage depends only on the throughput trace and calibrator, so the distribution is host-invariant by construction on this pool (RQ2).

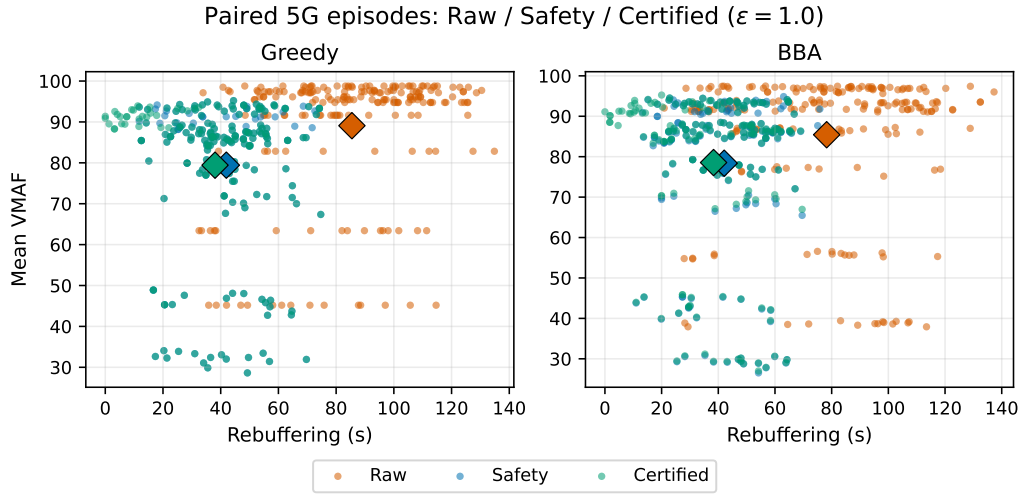


Figure 5: Per-episode operating points on synthetic 5G (paired seeds). Diamonds mark arm means. Safety removes catastrophic stalls relative to Raw; certified banking further reduces rebuffering at nearly unchanged mean VMAF.

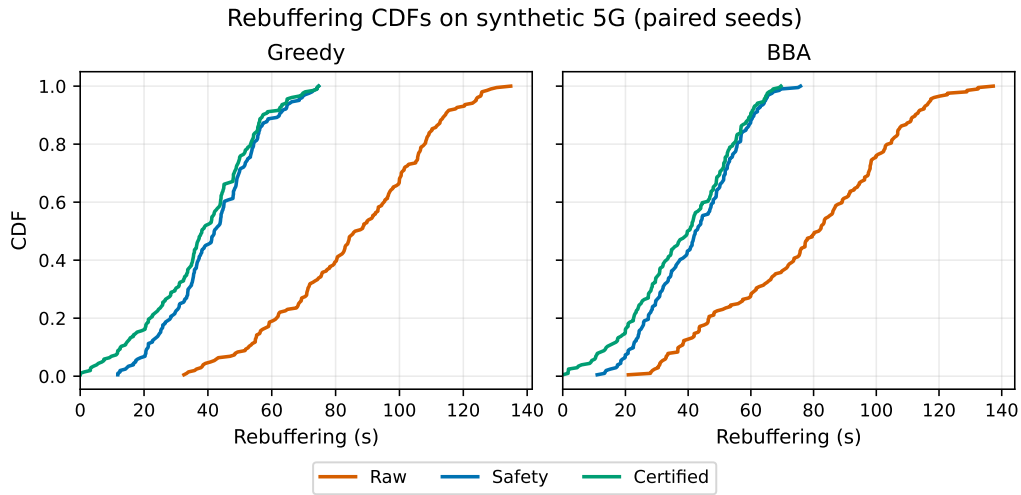


Figure 6: Empirical CDFs of episode rebuffering for greedy and BBA hosts. The certified arm stochastically dominates safety and raw on the stall axis.

Table 5: Paired arms on synthetic 5G test traces (204 episodes, $\varepsilon=1.0$, $\alpha=0.10$). Reb. is total rebuffering; Cover. is empirical conformal coverage; Interv. is shield intervention rate.

Host	Arm	Reb.	VMAF	Bitrate	Cover.	Interv.
Greedy	RAW	85.5 s	89.1	6000	—	0.0
	SAFETY	41.9 s	79.4	3131	0.919	65.8%
	CERTIFIED	38.0 s	79.4	2990	0.919	70.4%
BBA	RAW	78.2 s	85.4	4662	—	0.0
	SAFETY	42.1 s	78.4	2809	0.919	58.1%
	CERTIFIED	38.3 s	78.5	2692	0.919	66.1%
BOLA	RAW	77.6 s	83.2	4627	—	0.0
	SAFETY	42.4 s	77.5	2782	0.919	54.2%
	CERTIFIED	38.7 s	77.7	2667	0.919	61.1%
RobustMPC	RAW	71.3 s	77.4	3175	—	0.0
	SAFETY	40.0 s	74.9	2377	0.919	28.3%
	CERTIFIED	36.3 s	74.9	2287	0.919	31.1%
Pensieve	RAW	0.1 s	72.2	722	—	0.0
	SAFETY	0.1 s	70.8	689	0.919	7.3%
	CERTIFIED	0.1 s	70.8	689	0.919	7.4%

Certified–safety Wilcoxon: greedy bitrate $p=8.5\times 10^{-4}$, rebuffering $p=8.5\times 10^{-4}$; BBA bitrate $p=4.2\times 10^{-16}$, rebuffering $p=2.5\times 10^{-4}$. Improved predictive banking defaults in Table 4.

5.3. Predictive banking and broadband boundary (RQ3)

With risk-aware $\varepsilon(B)$ and dip forecasting enabled, the certified–safety bitrate reduction climbs to -14.6% (greedy) and -12.5% (BBA), and rebuffering falls -12.6% and -13.0% . Mean VMAF changes remain negligible (-0.23 , -0.22).

The broadband suite marks the edge of these gains. On source-disjoint real-measured broadband traces (FCC MBA and Norway/Nor-net [35, 36]), the greedy certified–safety bitrate change falls to -0.9% . Top rungs are seldom jointly feasible and saturated, so banking has little remaining headroom. Even so, safety vs. raw still cuts rebuffering by -98.7% . Under broadband stress the conformal feasibility layer—not banking—accounts for most of the improvement. On the same real-measured traces at $\alpha=0.10$, empirical conformal coverage is 0.890 (below the 0.90 target at $n=204$ episodes), whereas the synthetic 5G pool reaches 0.919 (Table 5, Fig. 4). The “BB greedy” group in Fig. 3 makes the regime split explicit: banking savings are largest on synthetic 5G. This split is by design: CPS banking pays off precisely when the ladder knee sits within reach, and broadband places it out of

reach.

Coverage under injected drift. To stress-test exchangeability without new traces, we replay the synthetic 5G pool with a mid-episode throughput shock (chunks 28–37: multiply trace throughput by 0.3, mimicking a sudden LOS/NLOS dip). After a 20-chunk conformal warm-up, per-chunk coverage in the steady pre-shock window is 0.928 ($n=1,632$ chunks). During the shock it is 0.929 ($n=2,040$); in the post-shock recovery window it rises to 0.944 ($n=2,040$), all against a $1-\alpha=0.90$ target (greedy certified arm, 204 episodes). Fig. 7 plots the rolling chunk coverage; the bound tracks the shock without collapsing below the target, consistent with the sliding-window defense in Section 4.1.

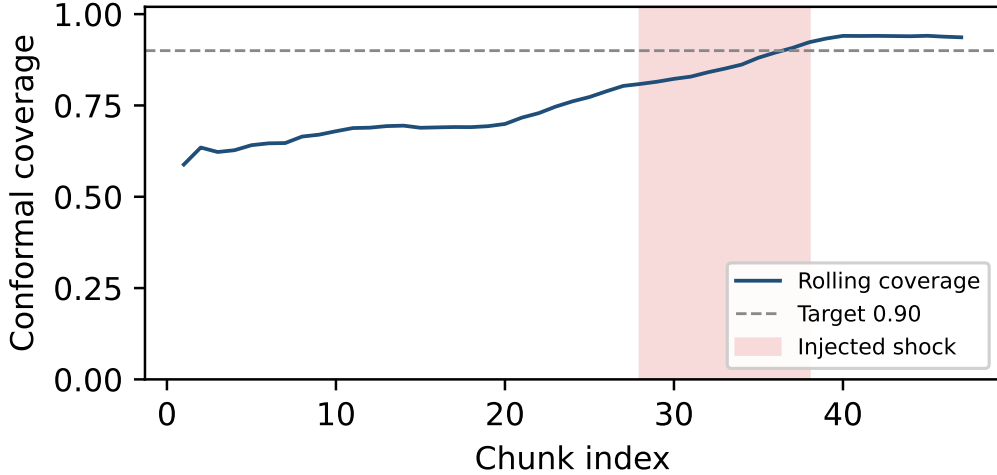


Figure 7: Rolling per-chunk conformal coverage under an injected throughput shock (shaded; $0.3\times$ scale, chunks 28–37) on the synthetic 5G pool. Dashed line: target $1-\alpha=0.90$.

5.4. Learned hosts and co-design (RQ4)

Two PPO settings warrant separate reporting. The primary host is a content-aware policy trained on source-disjoint broadband traces and evaluated, without further training, on 5G—the model-agnostic deployment case. The co-design pair comprises two policies trained on matched 5G traces—one shield-at-eval-only, one co-trained with CPS in the loop—compared on their certified arms (Table 6 and Fig. 8). On the primary broadband-trained policy, certified banking vs. safety returns -35.3% bitrate and -79.3% rebuffering ($p=6.2\times 10^{-18}$), at VMAF $\Delta = -0.33$ (TOST within $\pm\epsilon$). The effect exceeds that of greedy or BBA. The learned policy still proposes high rungs often on the twelve-title suite, which leaves

more saturation headroom for CPS. Under the improved shield, banking reaches $-21.9\%/ -29.6\%$.

Co-design. Both policies train on matched 5G traces (source-disjoint test). Under the improved shield at eval, the co-trained policy’s certified arm attains VMAF 75.73 and bitrate 1463 kb/s, against 72.80/801 for shield-at-eval-only operation. The **Δ row** in Table 6 is the difference of those certified-arm means ($+2.93$ VMAF, $+0.55$ s rebuffering), and it matches the raw rows above ($75.73 - 72.80$, $0.86 - 0.31$). The fidelity gain can be weighed against its stall cost through $\text{QoE}(w) = \overline{\text{VMAF}} - w \cdot \text{rebuffer}(\text{s})$. The mean-QoE crossover is $w^* = 5.31$ VMAF points per second (ratio of the Δ row entries): co-design favors fidelity-oriented weights $w < w^*$; shield-at-eval-only prevails where stall aversion is stronger. Paired Wilcoxon tests and episode-bootstrap intervals are reported as 5.0×10^{-5} (VMAF) and 5.0×10^{-5} (rebuffering). The bootstrap CI is $[2.61, 3.25]$, with stall cost $[0.29, 0.85]$ **s and crossover** $[3.40, 10.27]$ (median 5.37; positive VMAF in 100.0% of resamples). Co-design remains a supporting result under a single training seed (Section 6).

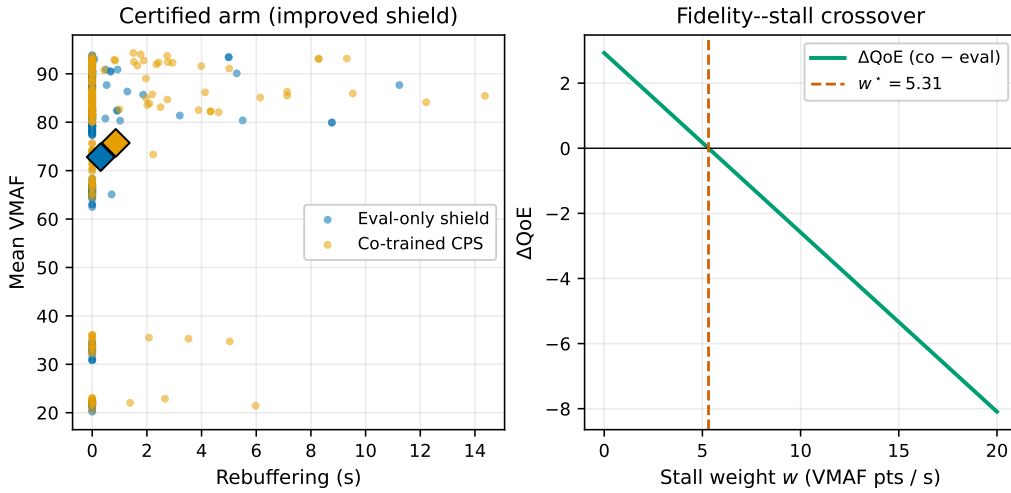


Figure 8: Co-design on 5G test (certified arm, improved shield). Left: per-episode VMAF–rebuffer points (diamonds = means). Right: mean $\Delta\text{QoE}(w)$ versus stall weight; the dashed line marks the crossover $w^* = 5.31$.

Table 6: Co-design on 5G test (certified arm, improved shield at eval).

Policy	VMAF	Reb. (s)	Bitrate	Cover.
Eval-only shield	72.80	0.31	801	0.919
Co-trained (CPS)	75.73	0.86	1463	0.919
Δ (co – eval)	+2.93	+0.55	—	—

The Δ row equals the certified-arm difference in the rows above. Paired Wilcoxon: VMAF $p=5.0\times 10^{-5}$, rebuffering $p=5.0\times 10^{-5}$; bootstrap Δ VMAF CI [2.61, 3.25] over 204 episodes (requires synced `episodes.csv`).

5.5. Per-chunk (non-monotone) ladder ablation (RQ5)

The headline runs apply the knee rule to the pooled session-mean ladder, which is monotone. We therefore test whether banking remains effective on the true per-chunk ladders, which are often non-monotone (26.5% of the 113 chunks hold at least one inverted adjacent pair; Table 3). To address this, we replay the deterministic banking stage on the per-chunk logs. We compare a shield that sees only the pooled ladder with one that sees the per-chunk ladder (Table 7, Fig. 9). Banking alone depends on VMAF; the feasibility projection ignores it and stays identical in both cases, which isolates the effect of the pooling simplification. Three findings hold at the default $\varepsilon=1.0$. First, the knee rule remains well defined on inverted ladders and banks more, not less: mean saved bitrate is 45.8% per-chunk against 44.6% pooled, because a lower rung that inverts above the proposal satisfies the budget immediately. Second, a per-chunk shield holds realized per-chunk perceptual cost inside budget by construction (mean 0.32 VMAF points, never above ε). Third, pooling is not free. Judged against the true per-chunk VMAF, the pooled-ladder shield overruns the budget on 11.5% of chunks (mean cost 0.33 points). Taken together, the pooled-ladder headline numbers are, if anything, conservative with respect to banking savings. Per-chunk operation is sound: it eliminates residual budget violations at the cost of a per-chunk VMAF feed, and it aligns the content-diversity motivation with the pooled evaluation.

5.6. Sensitivity to ε and α (RQ6)

The headline runs fix the pair at $\varepsilon=1.0$ and $\alpha=0.10$. Tables 8–9 and Fig. 10 replay the same greedy host on the synthetic 5G test pool at 204 paired episodes (twelve-title roster). We first consider the perceptual budget. Sweeping $\varepsilon \in \{0.5, \dots, 4.0\}$ (at $\alpha=0.10$) trades banking for fidelity. Bitrate saved versus safety climbs from 2.0% to 24.6%, while the mean VMAF change remains negligible (at most 0.48 ~~points~~) and never leaves the matching $\pm\varepsilon$ envelope. Empirical

Table 7: Banking-stage ablation on real per-chunk VMAF ladders (greedy proposal, 113 chunks). A per-chunk shield banks slightly more bytes than the pooled shield while keeping realized per-chunk cost $\leq \varepsilon$ by construction; the pooled shield incurs a small per-chunk budget-violation rate. Feasibility is VMAF-independent and unchanged.

ε	Mean knee rung		Bitrate saved (%)		Per-chunk cost (pts)	Budget viol. pooled (%)
	pooled	per-chunk	pooled	per-chunk	per-chunk	
0.5	4.58	4.34	22.3	28.1	0.02	8.8
1.0	4.15	3.94	44.6	45.8	0.32	11.5
2.0	3.60	3.39	58.2	59.7	0.98	18.6
4.0	2.93	2.88	68.7	68.7	1.89	7.1

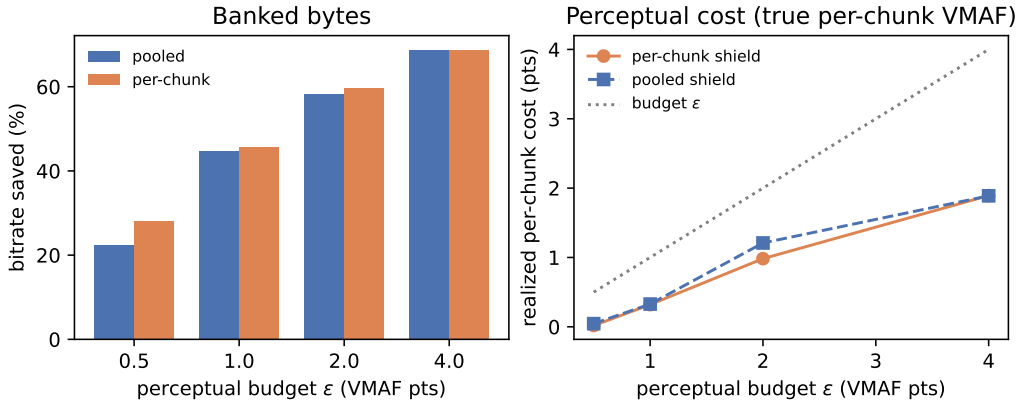


Figure 9: Per-chunk vs. pooled ladder banking across the perceptual budget ε . Left: banked bitrate is comparable (slightly larger per-chunk). Right: realized per-chunk perceptual cost stays under ε for the per-chunk shield; the pooled shield tracks it closely on average but violates the budget on a minority of chunks.

conformal coverage remains invariant with ε . Coverage depends on the throughput trace and calibrator alone, not the banking budget, which shows that banking and the feasibility certificate operate independently by design. We next consider miscoverage. Sweeping $\alpha \in \{0.05, 0.10, 0.20\}$ (at $\varepsilon=1.0$) on the same synthetic 5G pool moves empirical coverage in step with the finite-sample target $1-\alpha$ (coverage 0.945, 0.919, 0.854 for targets 0.95, 0.90, 0.80). Tighten α and both coverage and intervention rate rise; loosen it and both decrease, as the $(n+1)$ construction predicts. At $\alpha=0.05$ the mean is 0.945, marginally below the 0.95 target at $n=204$ (Section 5.3). The default operating point therefore lies on a smooth, well-behaved frontier rather than a narrowly tuned optimum.

Table 8: Sensitivity to the perceptual budget ε (greedy host, 5G test, $\alpha=0.10$, certified vs. safety). Banking scales smoothly with ε ; the mean VMAF change stays negligible and coverage is invariant to ε .

ε (pts)	Bitrate saved (%)	Rebuffer Δ (%)	VMAF Δ (pts)	Coverage
0.5	2.0	-3.5	+0.02	0.919
1.0	4.4	-8.2	-0.00	0.919
2.0	14.4	-30.8	-0.17	0.919
4.0	24.6	-46.1	-0.48	0.919

Table 9: Sensitivity to the conformal miscoverage α (greedy host, 5G test, $\varepsilon=1.0$). Empirical coverage tracks the finite-sample target $1-\alpha$, and the intervention rate rises as α tightens.

α	Target $1-\alpha$	Coverage	Interv. (%)	Bitrate saved (%)
0.05	0.95	0.945	73.0	4.5
0.10	0.90	0.919	71.0	4.4
0.20	0.80	0.854	35.2	6.2

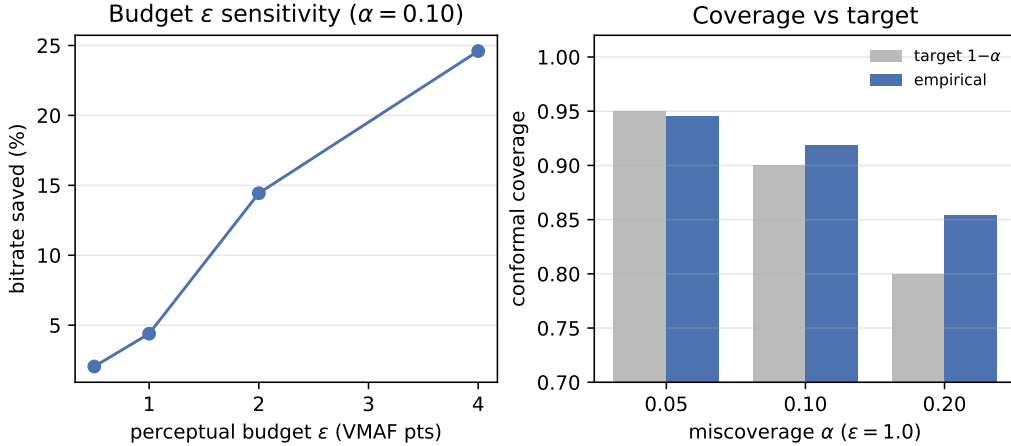


Figure 10: Hyperparameter sensitivity. Left: banked bitrate rises and the mean VMAF change stays near zero as the budget ε grows ($\alpha=0.10$). Right: empirical conformal coverage tracks the target $1-\alpha$ across α ($\varepsilon=1.0$).

6. Discussion and limitations

Scope. CPS is an auditable, model-agnostic mechanism that (i) banks bandwidth on VMAF-saturated ladder knees inside a stated perceptual budget and (ii) enforces download feasibility under a conformal throughput lower bound with

target coverage $1-\alpha$. These omissions are deliberate. The method does not claim dominance over every ABR controller, nor a pathwise worst-case guarantee beyond the conformal assumption of residual exchangeability. The payoff also depends on the regime. Banking gains accrue where high rungs are often feasible (moderate-buffer synthetic 5G). Under harsh broadband stress, the safety projection dominates and banking headroom decreases.

Local ladder inversions. Table 3 shows that, chunk by chunk, a sizable share of adjacent rungs in titles like Sintel and Big Buck Bunny run near-flat or inverted. A higher bitrate then scores no better, and sometimes worse, in VMAF. CPS handles all of this without a special case. Structurally, the evaluated shield runs on the monotone session-mean ladder, so Proposition 2 applies outright. Even on a fully per-chunk, non-monotone ladder, both stages remain well defined. Each stage, moreover, handles inversions on its own terms. The feasibility projection (6) uses download time, never VMAF, so inversions leave safety untouched. The knee rule (7) picks the smallest rung within ε of the proposal’s VMAF. When an inversion lets a lower rung score at least as high, that rung satisfies the budget on its own, and banking captures the extra headroom rather than failing on it. Local inversions, then, can only widen the banking opportunity, and the algorithm handles them by construction. The per-chunk ablation in Section 5.5 confirms this: banked bytes match, and realized per-chunk perceptual cost stays inside budget. Per-chunk ladders buy accuracy, not correctness.

Threats to validity.

1. The broadband suite replays real-measured FCC and Norway/Norinet traces [35, 36], yet the 5G suite is synthetic. In both cases the chunk-level emulator omits transport-stack dynamics and CDN caching [5]. Live-player validation is left to future work.
2. Twelve publicly available titles and finite trace pools still cap catalogue diversity [36]; three are held out from PPO training but recur in CPS evaluation.
3. QoE here is a VMAF-based surrogate. Subjective rankings of stall against fidelity may diverge from it [7].
4. Conformal coverage presumes exchangeable residuals (Section 4.1). At $n=204$ episodes, empirical coverage can sit marginally below the nominal $1-\alpha$ target on real broadband traces (0.890 at $\alpha=0.10$) or at tight miscoverage on synthetic 5G (0.945 vs. 0.95 at $\alpha=0.05$; Table 9). Headline RQ2 reports the default 5G operating point (0.919 vs. 0.90).
5. **Every arm—greedy, BBA, the PPO hosts—runs** locally under the same finite-sample $(n+1)$ conformal correction. Co-design magnitudes hinge on

the checkpoint, so we report the best-validation model of each policy. The ~~A-rows~~ of Table 6 come from certified-arm means. The paired Wilcoxon test and bootstrap intervals use per-episode exports over the same 204 pairs, with both co-design policies re-evaluated together. One point deserves emphasis: the co-design contrast uses a single training seed per policy, so the 95% intervals speak to evaluation variance, not training-seed variance. Establishing training-seed variance for w^* would require retraining many seeds, which we leave to future work. Hence co-design stands as a supporting result, not a headline claim.

6. Banking savings scale with ladder saturation (Table 3). The greedy top-gain split in Section 5.2 quantifies the steep-ladder penalty. The twelve-title mix deliberately includes such titles so pooled effect sizes are conservative relative to a saturation-rich subset alone.
7. The Pensieve-style no-op reveals that host aggressiveness sets the banking opportunity, so results should not be carried over to arbitrary policies without the paired-arm protocol.
8. The TOST margin equals the configured banking budget ε (Section 5.1); we justify $\varepsilon=1.0$ relative to ladder spacing and perceptual encoding practice [34], but it remains a design parameter rather than a clinical equivalence margin [41].

Future work. Several directions remain open: packet-level or field 5G evaluation; applying CPS to RobustMPC and Comyco hosts; CVaR/tail-aware objectives [42]; telemetry-driven adaptation of ε and α ; and preregistered non-inferiority studies with a fixed perceptual margin. Real-player deployment is the highest-priority direction. Every result here comes from a chunk-level emulator that omits transport-stack dynamics. Inserting CPS into a production player such as `dash.js` [40] therefore raises concrete challenges that the simulation never sees. Three challenges are central. First, throughput must be estimated from noisy, bursty HTTP/TCP (or QUIC) samples. Their exchangeability—the assumption behind the conformal bound—can break under slow-start, congestion-control transients, and CDN re-routing. The calibration residuals would then require periodic re-estimation, or a sliding, weighted-conformal variant that provably bounds coverage loss under drift [38]. Second, the feasibility check must reconcile the player’s own buffer-occupancy signal, which is segment-level and delayed, with the certificate’s buffer-margin accounting. Third, the per-decision budget tightens from our $\sim 100\mu\text{s}$ microbenchmark into a soft real-time constraint running beside the player’s ABR loop. Showing that the finite-sample coverage guarantee withstands

these real-world perturbations—and measuring whatever coverage slack they open up—is the natural next step.

7. Conclusion

We present the Certified Perceptual Shield (CPS) for client-side ABR. It comprises three components: VMAF-knee bandwidth banking, an online conformal throughput lower bound with finite-sample coverage, and certified buffer feasibility. Across 204 paired synthetic 5G episodes over twelve titles with diverse ladders, the results are consistent. Banking against conformal safety-only projection reduces both delivered bitrate and rebuffering across every deterministic host ($\approx -4.5\%/-9.3\%$ for greedy through $-3.8\%/-9.2\%$ for RobustMPC). Mean VMAF stays inside ± 1.0 , and empirical coverage is 0.919 against a 0.90 target. Predictive banking further improves the margin (-14.6% greedy bitrate). Under broadband stress, safety alone predominates once the ladder knee lies beyond the observable operating range. As a supporting single-seed result, optional PPO co-training with CPS in the loop is reported in Section 5.4 ($+2.93$ VMAF at $+0.55$ s rebuffering; crossover $w^*=5.31$). CPS integrates with existing ABR controllers as a lightweight wrapper—no retraining required—and extends naturally to learned policies.

Appendix A. Lagrangian CMDP training for RL hosts

An unconstrained MDP chases $\mathbb{E}[\sum_k \gamma^k r_k]$ and nothing else—no explicit budget on stalls or smoothness. So when the base policy is learned, we may recast streaming as a constrained MDP (CMDP) that maximizes expected fidelity under budgets on trajectory-level auxiliary costs [29]. Set $c_k^{\text{stall}} \equiv \tau_k^{\text{stall}}$ and $c_k^{\text{sm}} \equiv \text{smooth}_k$. The undiscounted episodic rebuffering ratio and mean smoothness cost read

$$\text{RB}(e) = \frac{1}{T_e^{\text{play}}} \sum_{k \in \mathcal{K}_e} c_k^{\text{stall}}, \quad \text{SM}(e) = \frac{1}{K_e} \sum_{k \in \mathcal{K}_e} c_k^{\text{sm}}, \quad (\text{A.1})$$

with targets $\delta_{\text{rf}}=0.05$ and $\delta_{\text{sm}}=3.5$ [2]. Per-step normalized costs $\widetilde{\text{RB}}_k$ and $\widetilde{\text{SM}}_k$ yield the Lagrangian-relaxed PPO reward

$$r_k^{\text{train}} = \frac{1}{100} \left(q_k - \lambda_r \widetilde{\text{RB}}_k - \lambda_s \widetilde{\text{SM}}_k - w_b \phi(B_k) - w_\ell \psi_k^{\text{Lya}} \right), \quad (\text{A.2})$$

where projected ascent on the empirical constraint gaps updates the dual variables (λ_r, λ_s) [30]. Training may also subtract buffer-deviation and Lyapunov drift

penalties [37]; these steady the optimization but certify nothing. The gap is real: finite-sample PPO does not certify per-chunk feasibility [4, 31], and CPS hands over that layer at deployment time.

The implementation rides Stable-Baselines 3 PPO [4, 39]: two Tanh MLPs (width 256), GAE ($\gamma=0.99$, $\lambda_{\text{GAE}}=0.95$), dual step sizes $\eta_r=0.012$ and $\eta_s=0.003$, and projected ranges $\lambda_r \in [4.3, 40]$, $\lambda_s \in [0.3, 2.5]$ after a 60-episode warm-up. As ~~in the main text~~, co-design wraps the training environment in CPS; shield-at-eval-only policies train without it.

Data availability

The code, evaluation scripts, and exported CSV files underlying Tables 5–6 and Figures 3–8 are available with the article artifacts. The Certified Perceptual Shield implementation and its parameters are documented with the code.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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